

Ques

Find area of the figure enclosed by
 $5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$.

Sol

$$2y^2 + 6(x+1)y + 5x^2 + 7x + 6 = 0.$$

$$y = \frac{-6(x+1) \pm \sqrt{36(x^2+2x+1) - 8(5x^2+7x+6)}}{2 \cdot 2}$$

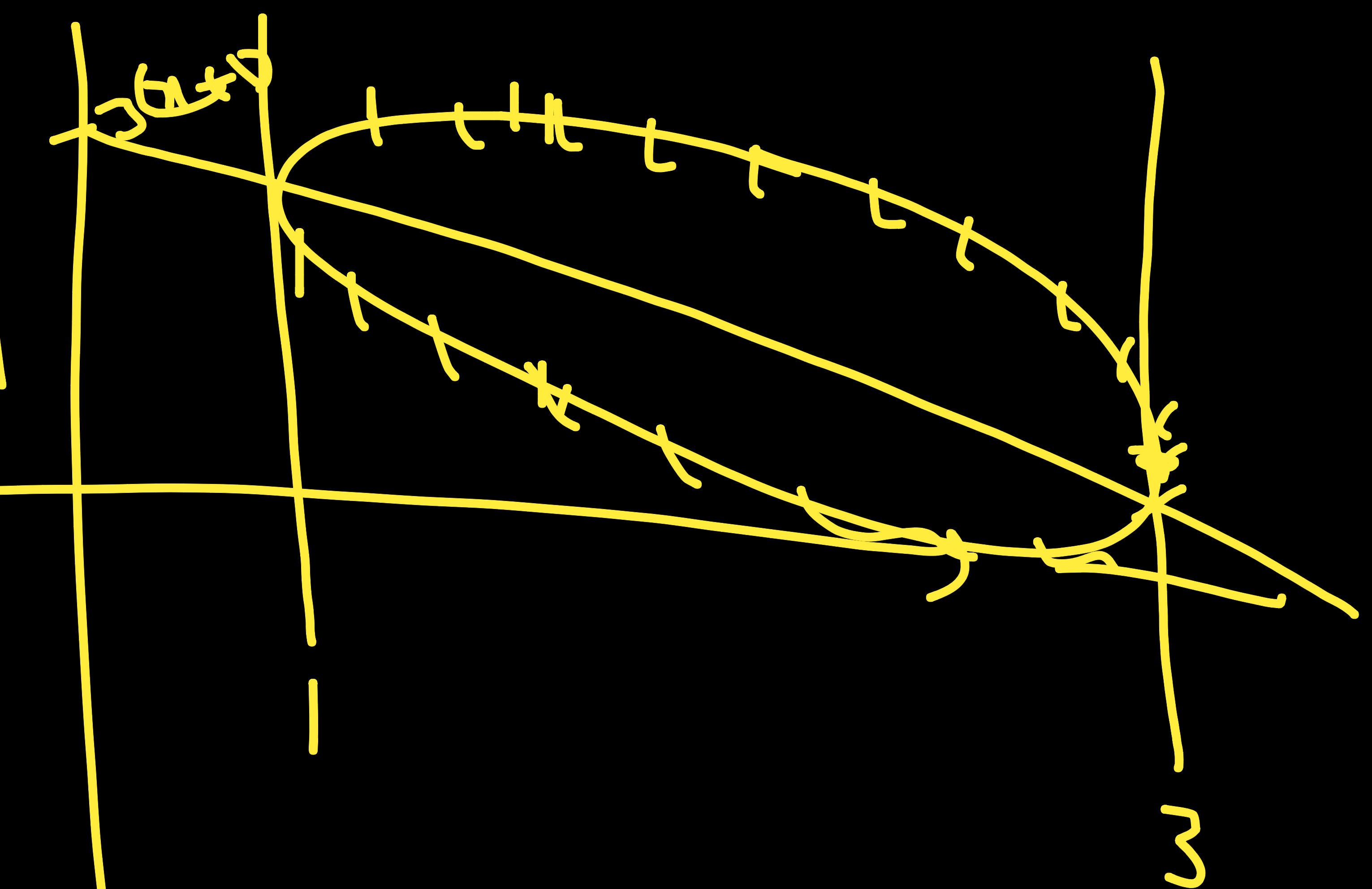
$$= \frac{-3(x+1) \pm \sqrt{-x^2 + 4x - 3}}{2} = \frac{-3(x+1) \pm \sqrt{-(x^2 - 4x + 3)}}{2}$$

$$y = \frac{-3(x+1) + \sqrt{(x-1)(3-x)}}{2}$$

$$\textcircled{2} \quad y = \frac{-3(x+1) - \sqrt{(x-1)(3-x)}}{2}$$

$$x^2 - 4x + 3 \leq 0 \\ \Rightarrow x \in [1, 3]$$



$$R. Area = \int_1^3 \left(\frac{-3(x+1) + \sqrt{-x}}{2} - \left(\frac{-3(x+1) - \sqrt{-x}}{2} \right) \right) dx$$


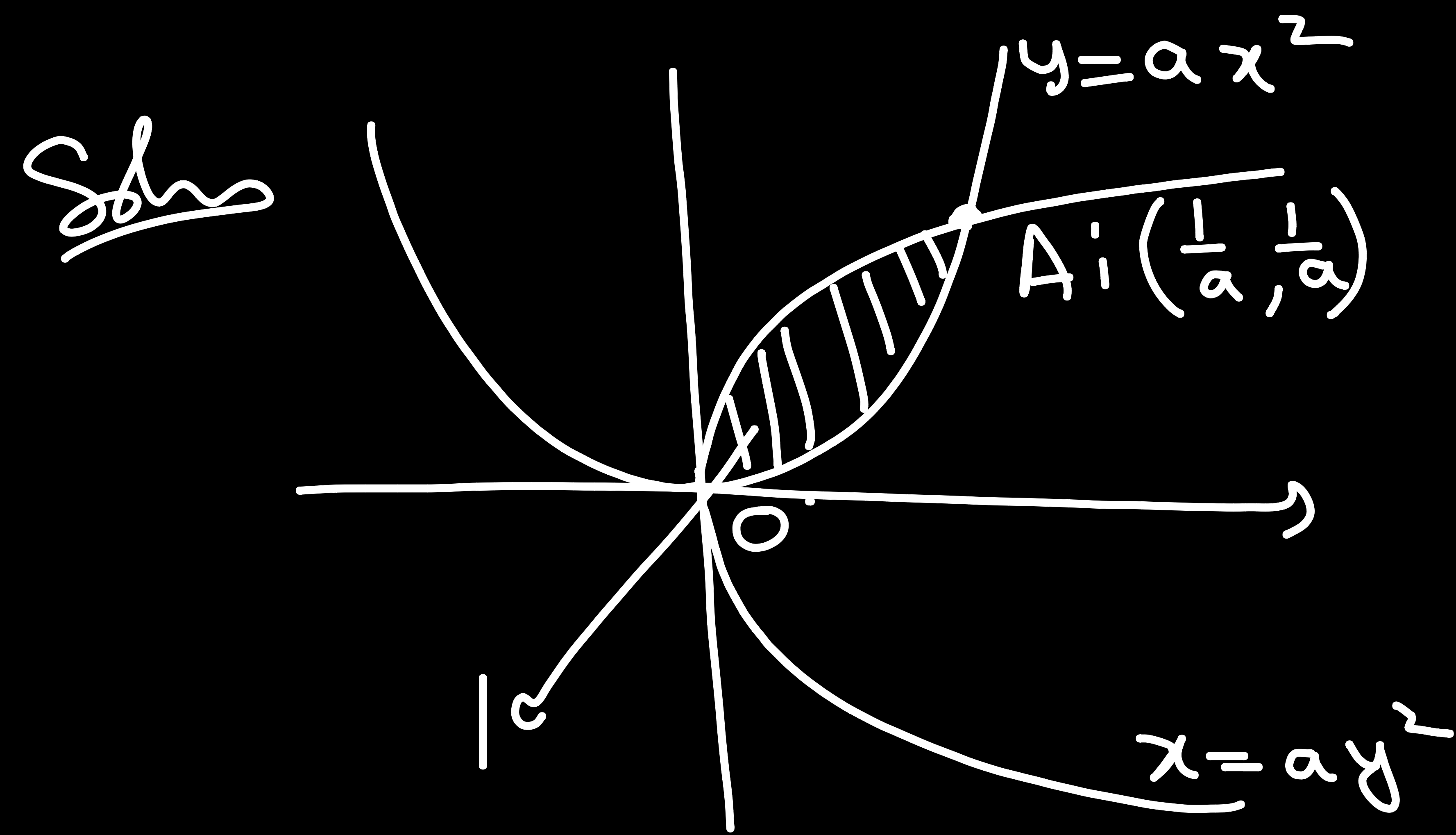
$$= \int_1^3 \sqrt{x-1} \sqrt{3-x} \, dx$$

$$= \frac{1}{2} \int_0^{\pi/2} \sqrt{2s^2} \cdot \sqrt{2c^2} \cdot \underline{2s \cdot c} \cdot d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \sin^2(2\theta) \cdot d\theta = \frac{1}{4} \int_0^{\pi/2} (1 - \cos(4\theta)) d\theta = \frac{1}{4} \left(\frac{\pi}{2} - \left[\frac{\sin(4\theta)}{4} \right]_0^{\pi/2} \right) = \frac{\pi}{8}$$

$$\begin{aligned} x &= 3\sin^2(\theta) + \cos^2(\theta) \rightarrow x-1 = 2\sin^2(\theta) \\ dx &= 2(3s \cdot c - s \cdot c) d\theta = 2(2s \cdot c \cdot d\theta) = 2\sin(2\theta) d\theta \\ x=1 &\rightarrow s^2 = 3s^2 \Rightarrow \theta = 0 \\ x=3 &\rightarrow 3c^2 = c^2 \Rightarrow \theta = \pi/2 \end{aligned}$$

Que If area enclosed by $y = ax^2$ and $x = ay^2$ ($a > 0$) is 1-square unit then find 'a'.



$$\text{Area} = 1 = \int_0^{1/a} \left(\sqrt{\frac{x}{a}} - ax^2 \right) dx$$

$$1 = \left[\frac{2}{3} \frac{x^{3/2}}{\sqrt{a}} \right]_0^{1/a} - \frac{a}{3} [x^3]_0^{1/a}$$

$$1 = \frac{2}{3} \frac{1}{a^2} - \frac{1}{3} \cdot \frac{1}{a^3} a^2 = \frac{1}{3a^2}$$

$$a^2 = \frac{1}{3} \Rightarrow \boxed{a = \frac{1}{\sqrt{3}}} \quad \text{Ans}$$

for A

$$y = a(ay^2)^2$$

$$y = a^3 y^4 \Rightarrow y = \frac{1}{a}$$

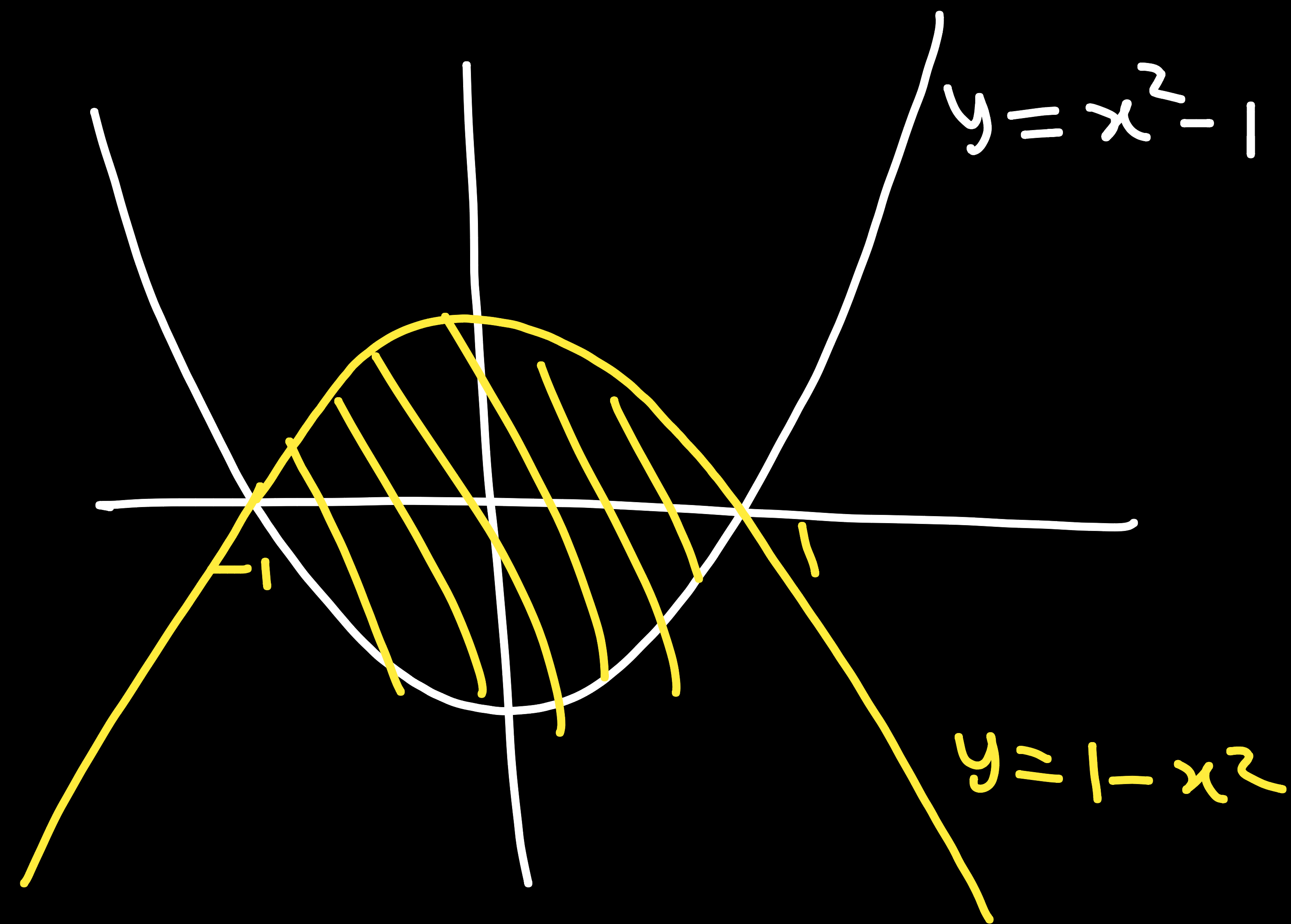
Ques Find the area enclosed by $x^4 + 1 = 2x^2 + y^2$.

Sol

$$x^4 - 2x^2 + 1 = y^2$$

$$(x^2 - 1)^2 = y^2$$

$$\Rightarrow y = \pm(x^2 - 1)$$



$$R.A. = 2 \int_{-1}^1 (1 - x^2) dx$$

$$= 4 \int_0^1 (1 - x^2) dx$$

$$= 4 \left[x - \frac{x^3}{3} \right]_0^1 = 4 \left(1 - \frac{1}{3} \right) = \frac{8}{3}$$

Ans

Ques Let $T_n = \sum_{2n}^{3n-1} \frac{x}{x^2+n^2}$ (8) $S_n = \sum_{2n+1}^{3n} \frac{x}{x^2+n^2}$, for $n \in \mathbb{N}$,

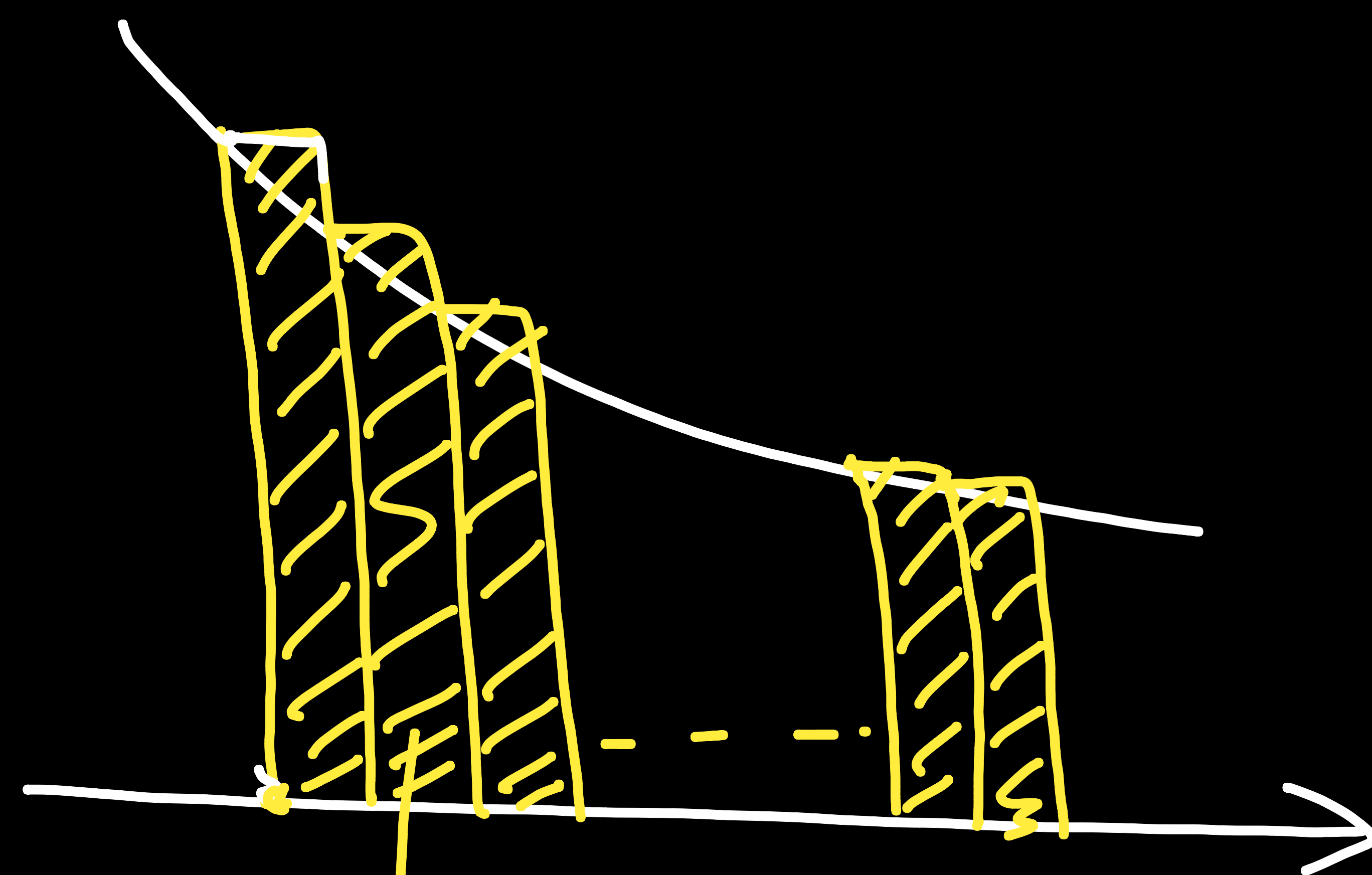
then identify correct statement (8):

(A) $T_n > \frac{1}{2} \ln(2)$

(B) $S_n < \frac{1}{2} \ln(2)$

(C) $T_n < \frac{1}{2} \ln(2)$ (D) $S_n > \frac{1}{2} \ln(2)$

Sol



Area = T_n

(A), (B)

$S_n < I < T_n$

$$\lim_{n \rightarrow \infty} \sum_{2n}^{3n-1} \frac{x}{x^2+n^2} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{2n}^{3n-1} \frac{x/n}{(x/n)^2+1}$$

$$= \int_2^3 \left(\frac{x}{x^2+1} \right) dx$$

$$= \left[\frac{1}{2} \ln(x^2+1) \right]_2^3$$

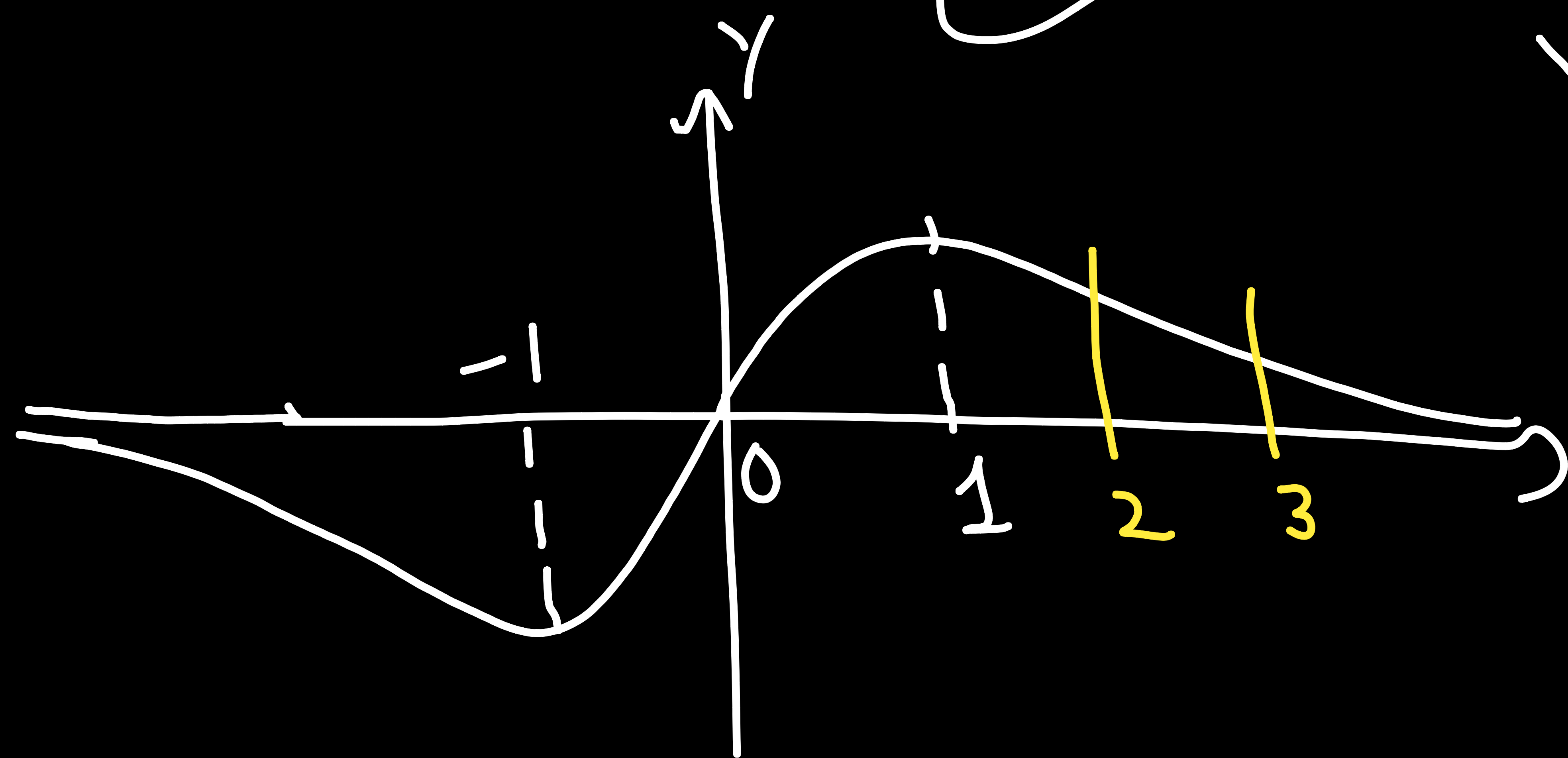
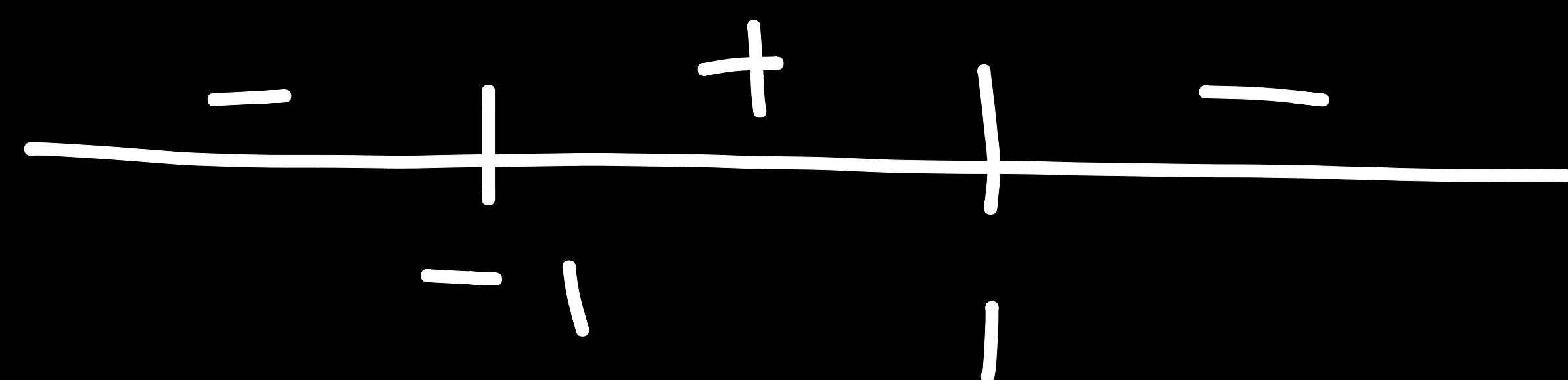
$$= \frac{1}{2} \ln\left(\frac{10}{5}\right) = \frac{1}{2} \ln(2)$$

$$f(x) = \frac{x}{x^2+1}$$

$$f'(x) = \frac{x^2+1-2x^2}{(x^2+1)^2}$$

$$= \frac{1-x^2}{(x^2+1)^2}$$

sign of f'



Ques Find area of the smaller region bounded by $\frac{(x-1)^2}{4} + (y-1)^2 = 1$

and $(y-1)^2 = \frac{3}{4}(x-1)$.

Sol $\frac{(x-1)^2}{4} + (y-1)^2 = 1$

$$(y-1)^2 = \frac{3}{4}(x-1)$$

↓

$$\frac{x^2}{4} + y^2 = 1$$

$$y^2 = \frac{3}{4}x$$

$$\Rightarrow \frac{x^2}{4} + \frac{3}{4}x = 1$$

$$x=1$$



$$R.A. = 2 \left(\int_0^1 \frac{\sqrt{3}}{2} \sqrt{x} \, dx + \int_1^2 \sqrt{1 - \frac{x^2}{4}} \, dx \right)$$

Ans

Que Let S be the area bounded by $y = e^{-x^2}$, $y = 0$, $x = 0$ and $x = 1$, then identify correct statement(s).

(A) $S \geq \frac{1}{e}$ (B) $S \geq 1 - \frac{1}{e}$ (C) $S \leq \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}}\right)$ (D) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}}\right)$

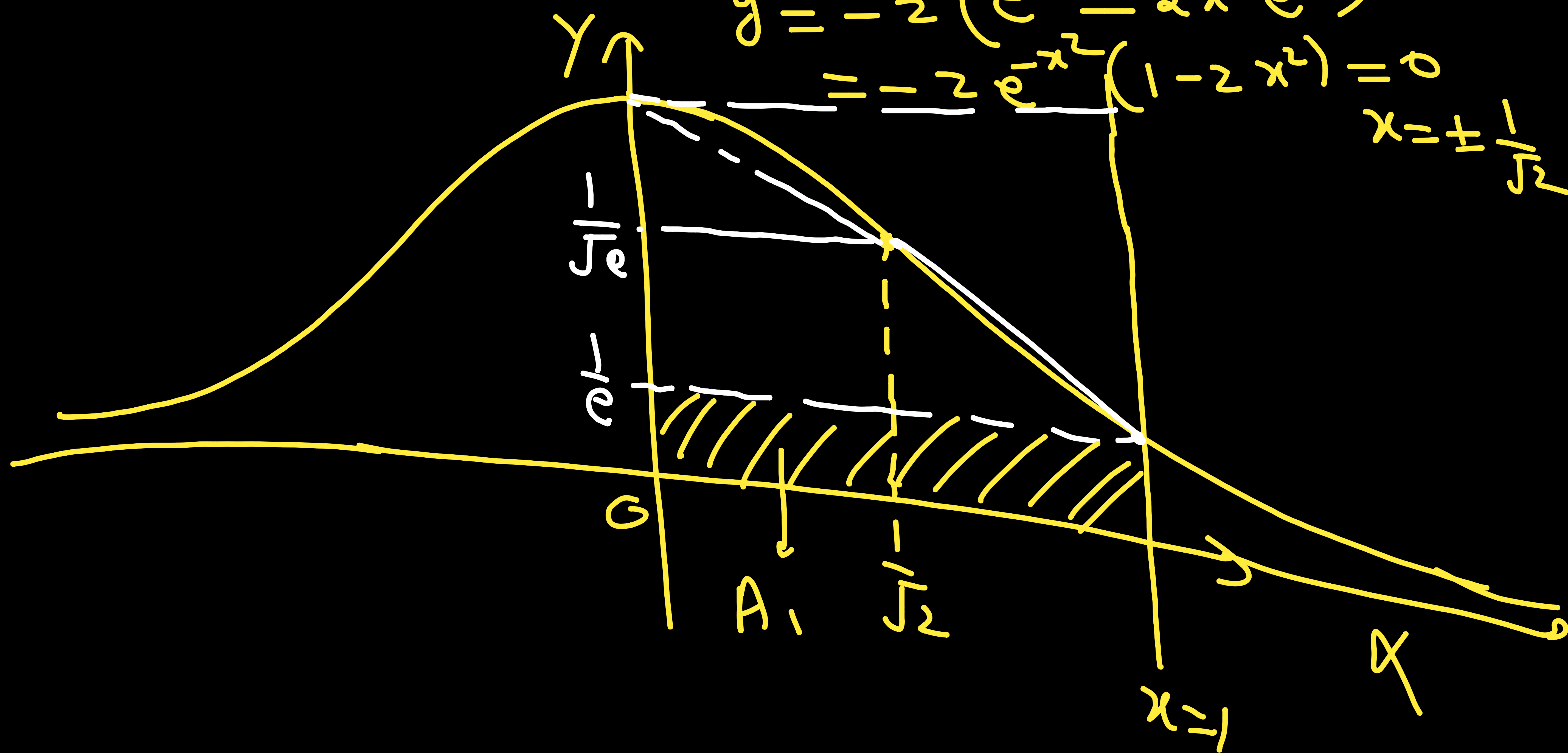
Sol

$$y = e^{-x^2} \Rightarrow y' = -2x e^{-x^2}$$

$$y'' = -2(e^{-x^2} - 2x^2 e^{-x^2})$$

$$= -2e^{-x^2}(1 - 2x^2) = 0$$

$$x = \pm \frac{1}{\sqrt{2}}$$



Clearly

$$S > A_1 = 1 \cdot \frac{1}{e}$$

$$S > \frac{1}{e} \quad \text{(A)}$$

$\ln(0,1) \quad e^x > e^{-x^2}$

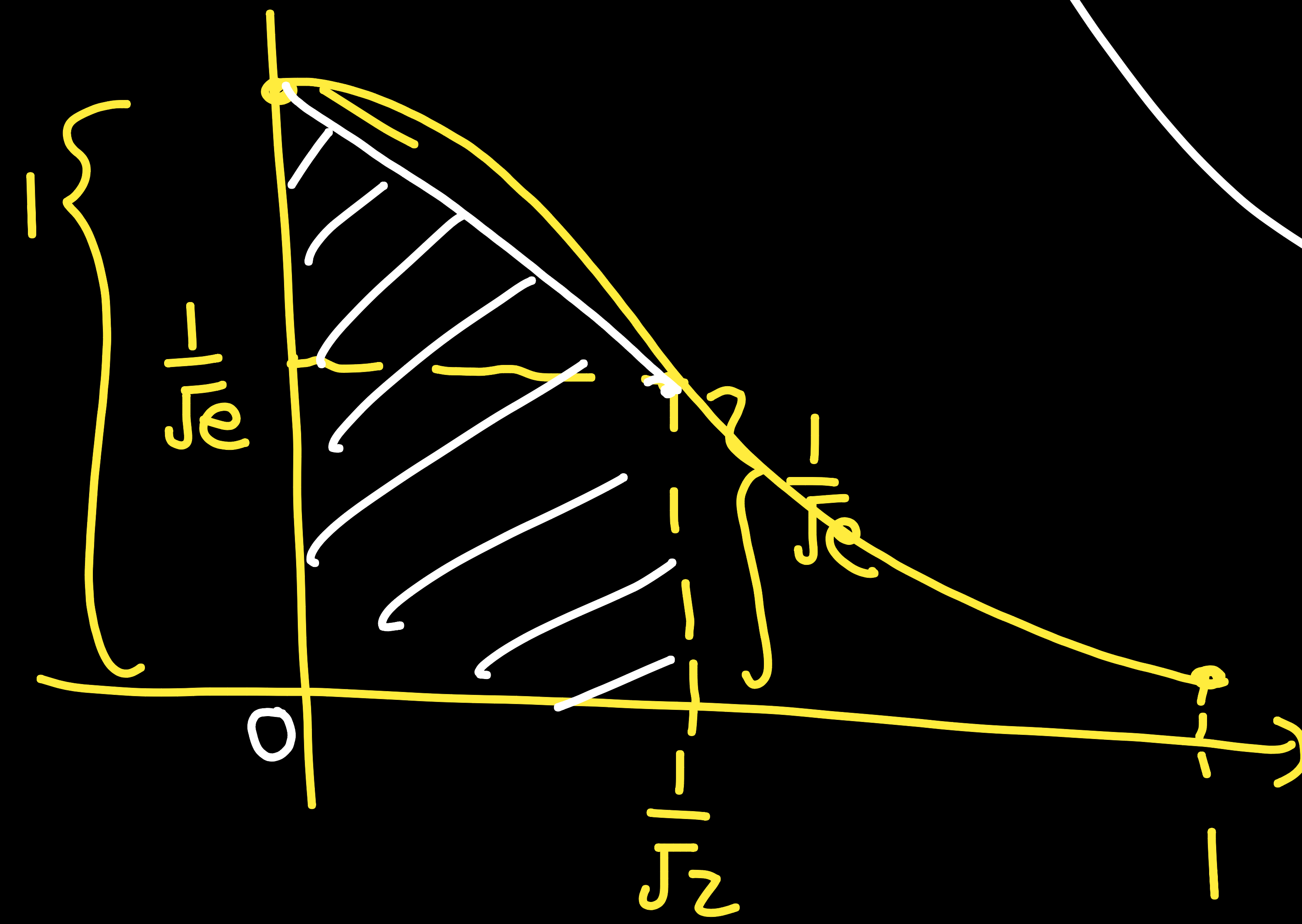
$$e^{-x} < e^{-x^2}$$

$$\int_0^1 e^{-x} dx < \int_0^1 e^{-x^2} dx$$

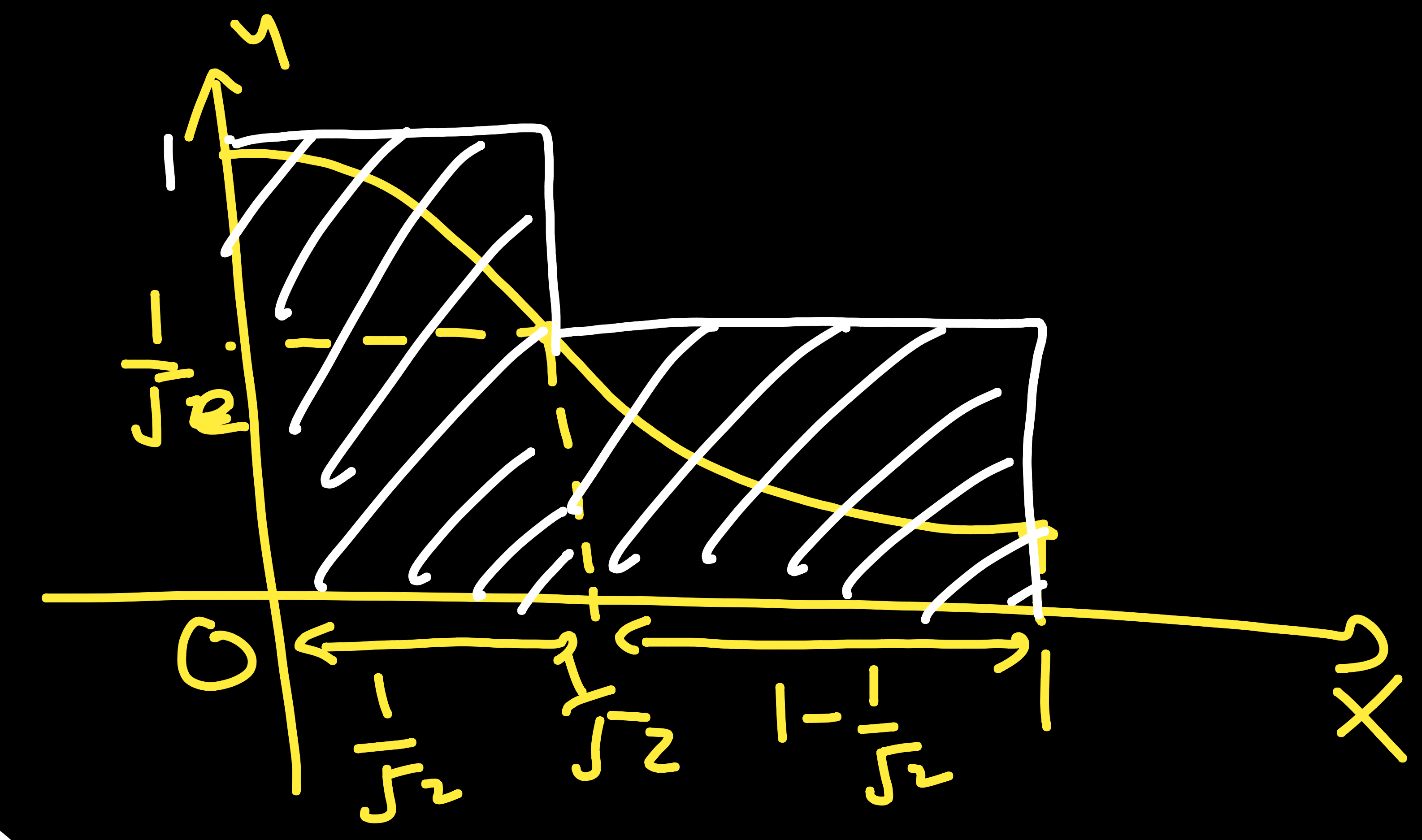
$$\left[-e^{-x} \right]_0^1 = 1 - \frac{1}{e}$$

$$\Rightarrow S > 1 - \frac{1}{e} \quad \text{(B)}$$

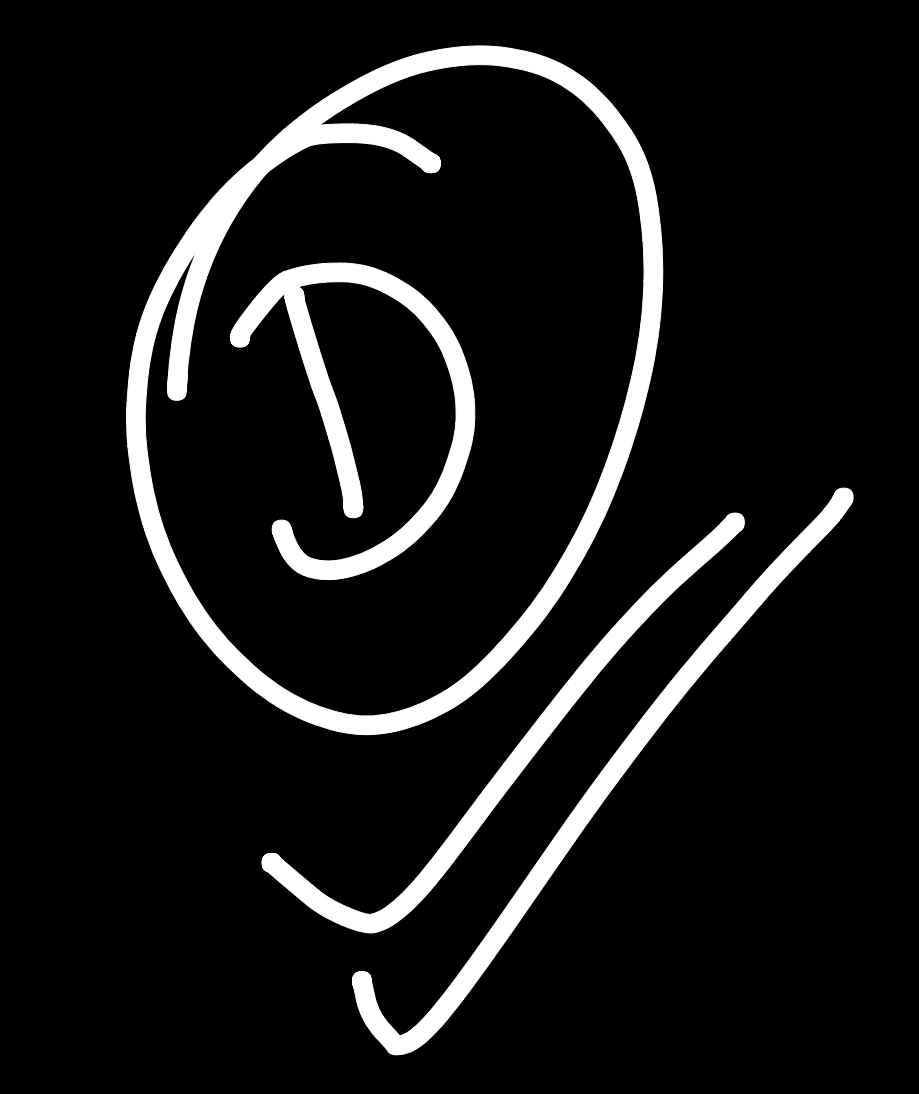
(C) $S \leq \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}}\right)$
 (D) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}}\right)$



$S > \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \cdot \left(\frac{1}{\sqrt{e}} + 1\right) > \frac{1}{2} \cdot \frac{1}{2} \left(\frac{1}{\sqrt{e}} + 1\right)$
 $= \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}}\right)$



$S < 1 \cdot \frac{1}{\sqrt{2}} + \left(1 - \frac{1}{\sqrt{2}}\right) \cdot \frac{1}{\sqrt{e}}$



(A) (B) (D)

Que

Find area of the region in which $P(x, y)$ can move while satisfying $x^2 + y^2 \leq 100$ and $2n \leq x+y \leq 2n+1, \forall n \in I.$

Sol

$$x^2 + y^2 \leq 100$$

$$2n \leq x+y \leq 2n+1$$

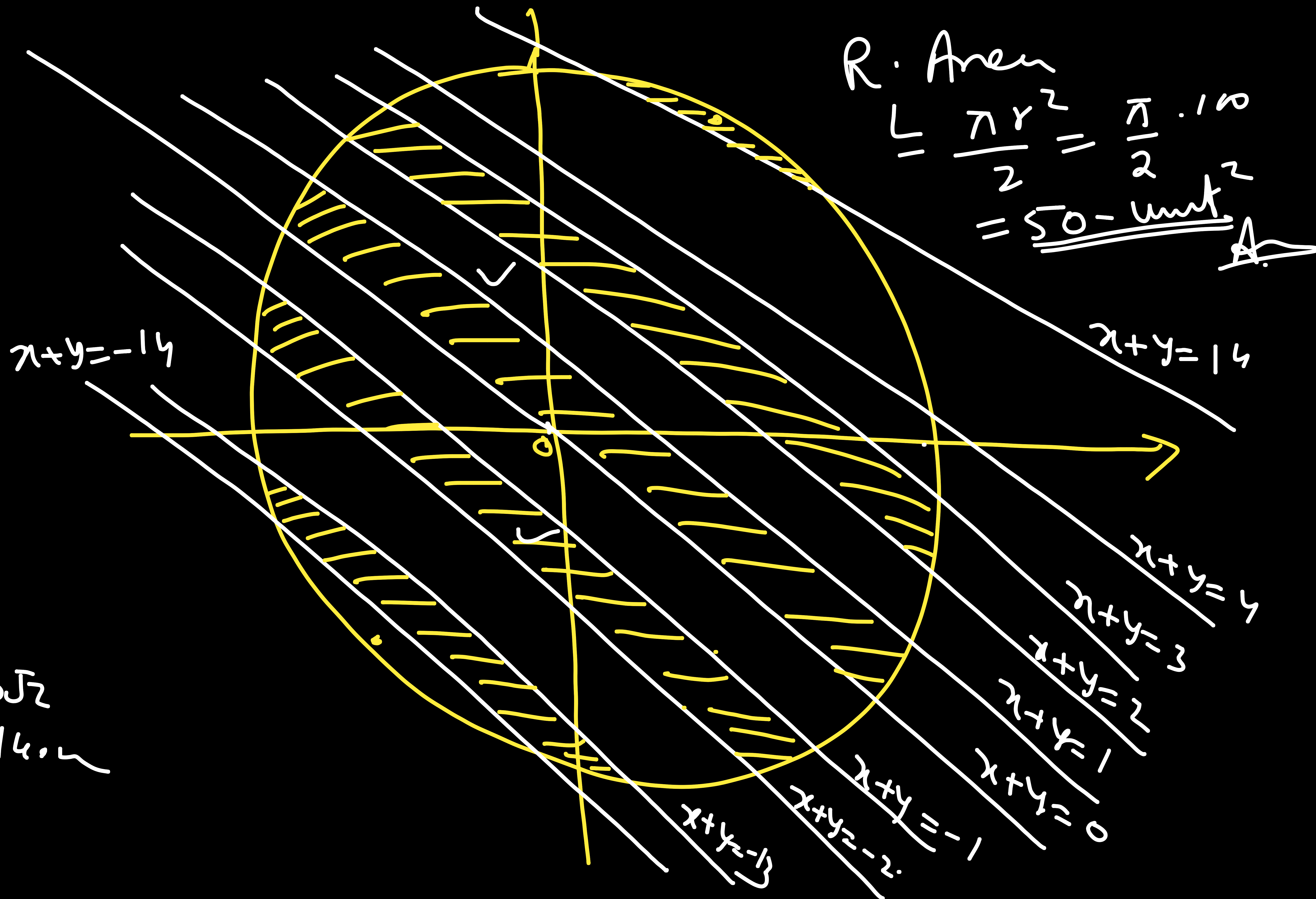
$$0 \leq x+y \leq 1$$

$$2 \leq x+y \leq 3$$

$$-2 \leq x+y \leq -1$$

$$x+y = n$$

$$\frac{n}{\sqrt{2}} = 10 \Rightarrow n = 10\sqrt{2} = 14.14$$



R. Area

$$= \frac{\pi r^2}{2} = \frac{\pi \cdot 100}{2} = 50\pi \text{ unit}^2$$

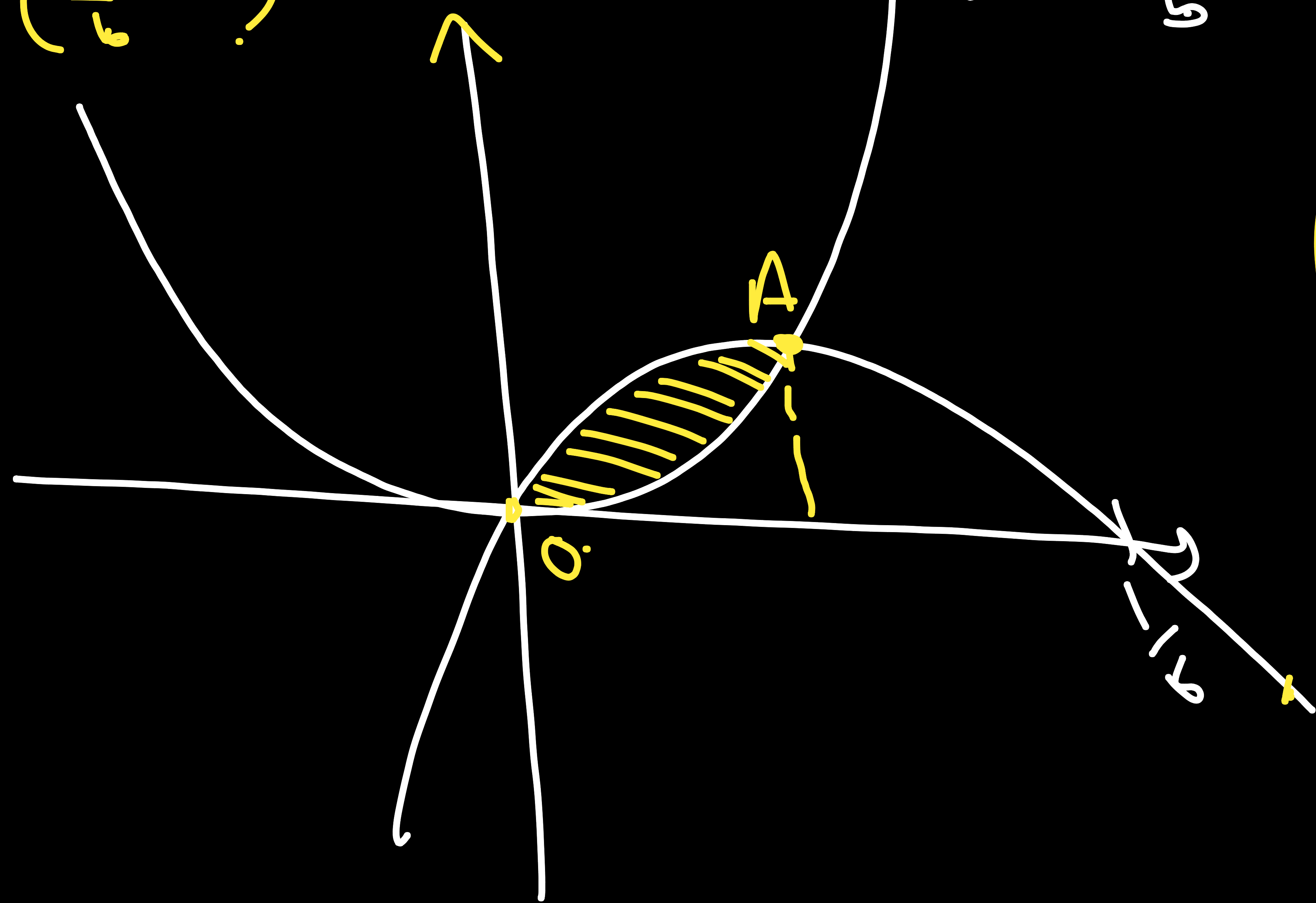
Ans

Que Find $b \in \mathbb{R}^+$ if the area bounded by $y = x - bx^2$ & $y = \frac{x^2}{b}$ is maximum.

Sol

$$\begin{aligned} y &= x - bx^2 \\ &= x(1 - bx) \\ &= bx\left(\frac{1}{b} - x\right) \end{aligned}$$

$$y = \frac{x^2}{b}$$



for A

$$x - bx^2 = \frac{x^2}{b}$$

$$1 - bx = \frac{x}{b} \Rightarrow x\left(b + \frac{1}{b}\right) = 1$$

$$x = \frac{b}{b^2 + 1}$$

$$y = \frac{x^2}{b}$$

$$\text{Area} = A(b) = \int_0^{b_1} \left((x - bx^2) - \frac{x^2}{b} \right) dx.$$

$$A'(b) = 0$$

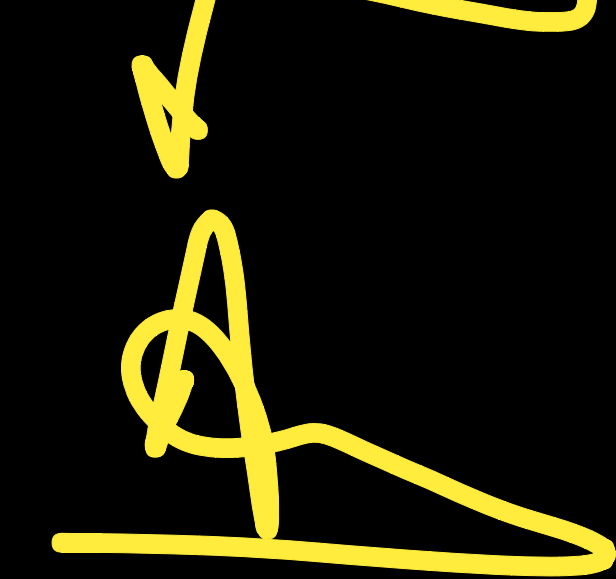
$$b_1' \left(b_1 - b b_1^2 - \frac{b_1^2}{b} \right) + \int_0^{b_1} \left(-x^2 + \frac{x^2}{b^2} \right) dx = 0.$$

$$\int_0^{b_1} x^2 \left(\frac{1}{b^2} - 1 \right) dx = 0.$$

$$\frac{x^3}{3} \left(\frac{1}{b^2} - 1 \right) = 0 \Rightarrow$$

$$\frac{1}{3} \left(\frac{1}{b^2} - 1 \right) = 0$$

$$\boxed{b = 1}$$

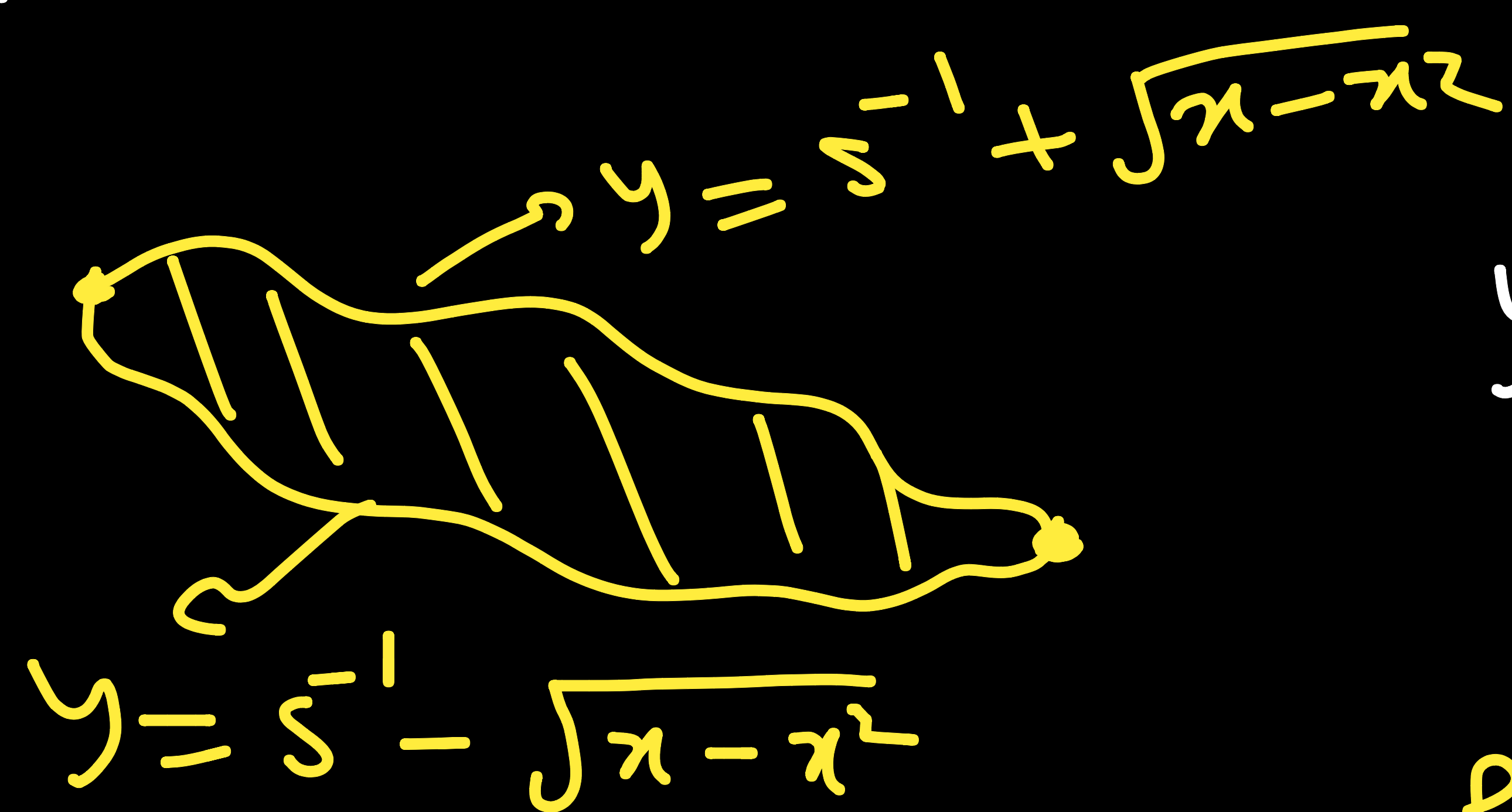


Ques Find area enclosed by $(y - \sin^{-1}(x))^2 = x - x^2$.

Sol

$$(y - \sin^{-1}(x))^2 = x - x^2 \Rightarrow y - \sin^{-1}(x) = \pm \sqrt{x - x^2}$$

$x \in [0, 1]$



$$y = \sin^{-1}(x) + \sqrt{x - x^2}$$

$$y = \sin^{-1}(x) - \sqrt{x - x^2}$$

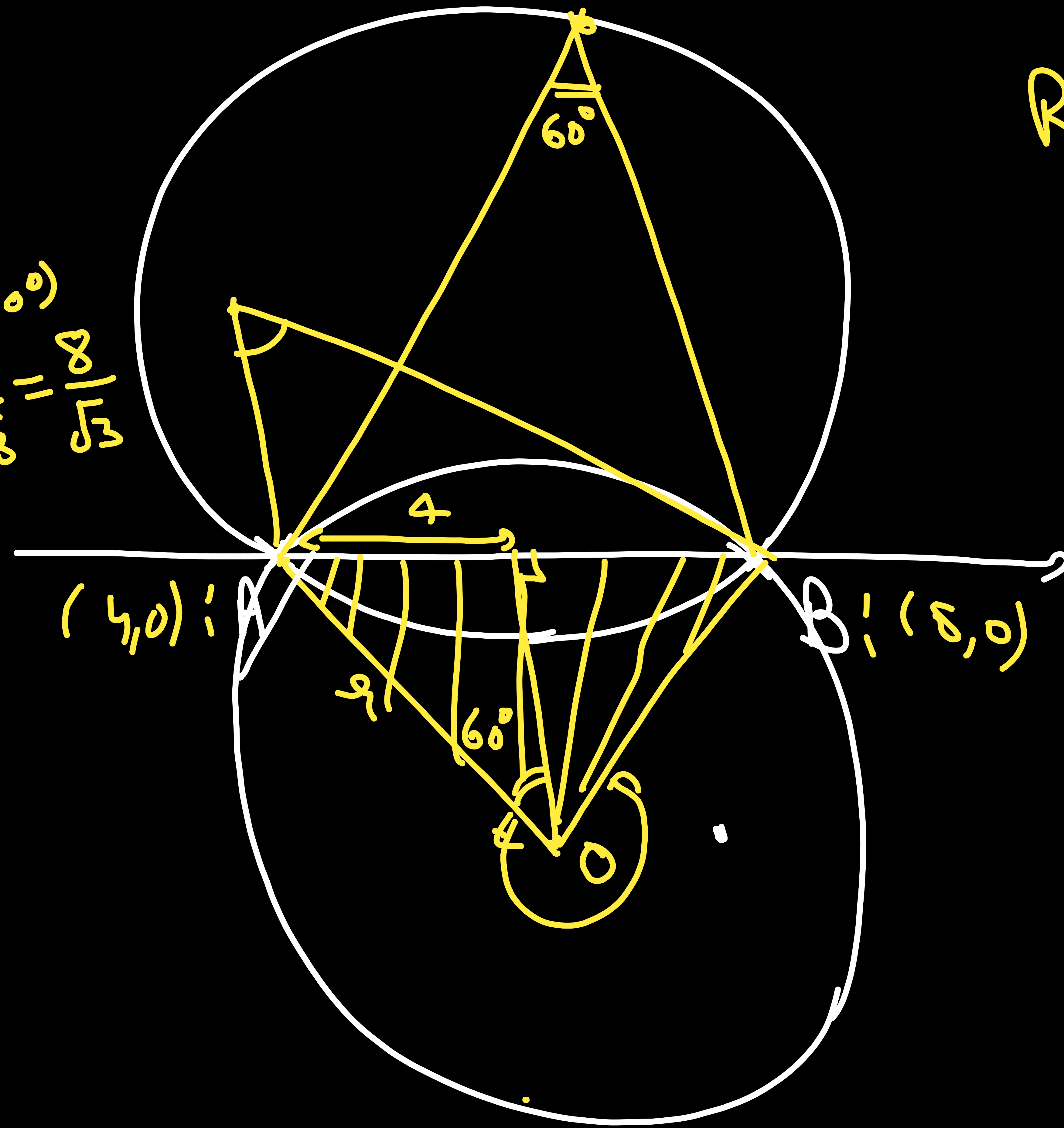
$$R.A. = \int_0^1 \left((\sin^{-1} + \sqrt{x - x^2}) - (\sin^{-1} - \sqrt{x - x^2}) \right) dx$$

$$= 2 \int_0^1 \sqrt{x - x^2} dx = \pi$$

Ques Let $A(4,0)$ and $B(8,0)$, if P is a variable point such that $\angle APB > 60^\circ$ then find area of the region in which P can move.

Sol

$$r = 4 \csc(60^\circ) \\ = 4 \cdot \frac{2}{\sqrt{3}} = \frac{8}{\sqrt{3}}$$



$$R.A. = 2 \cdot \left(\frac{1}{2} \cdot r^2 \cdot \sin(120^\circ) + \frac{1}{2} \cdot r^2 \cdot \frac{4\pi}{3} \right) \\ = \frac{64}{3} \left(\frac{\sqrt{3}}{2} + \frac{4\pi}{3} \right) \quad \underline{\underline{\text{Unit}^2}}$$

Ques If the area bounded by $y = x^2 + 2x - 3$ and the line $y = kx + 1$ is least, then find 'k' and the corresponding area.

for x_1, x_2

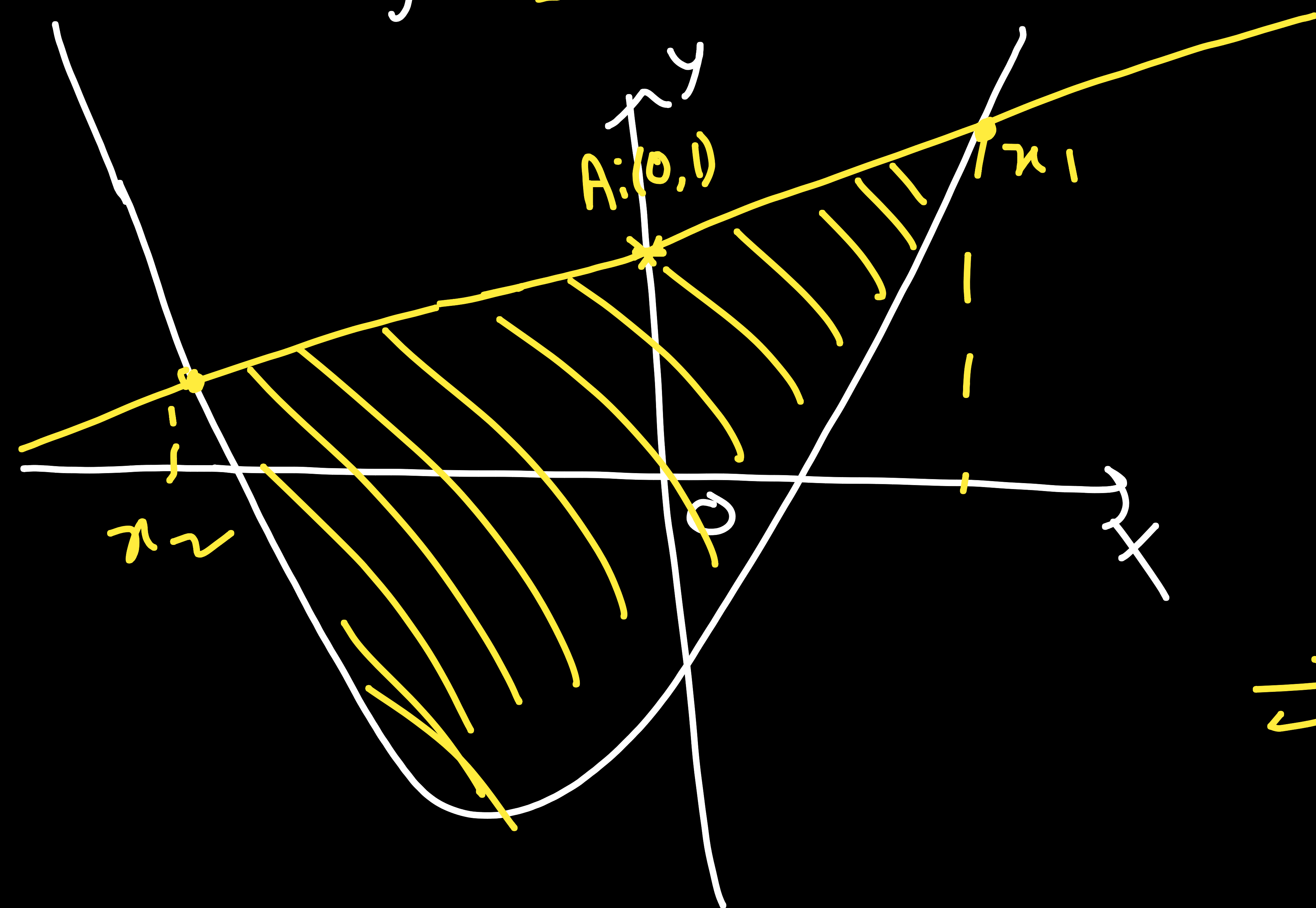
$$x^2 + 2x - 3 = kx + 1 \rightarrow x_1, x_2.$$

$$\Rightarrow k - 2 = x_1 + x_2$$

Sol

$$y = x^2 + 2x - 3$$

$$y = kx + 1$$



$$A(k) = \int_{x_2}^{x_1} (kx + 1 - (x^2 + 2x - 3)) dx$$

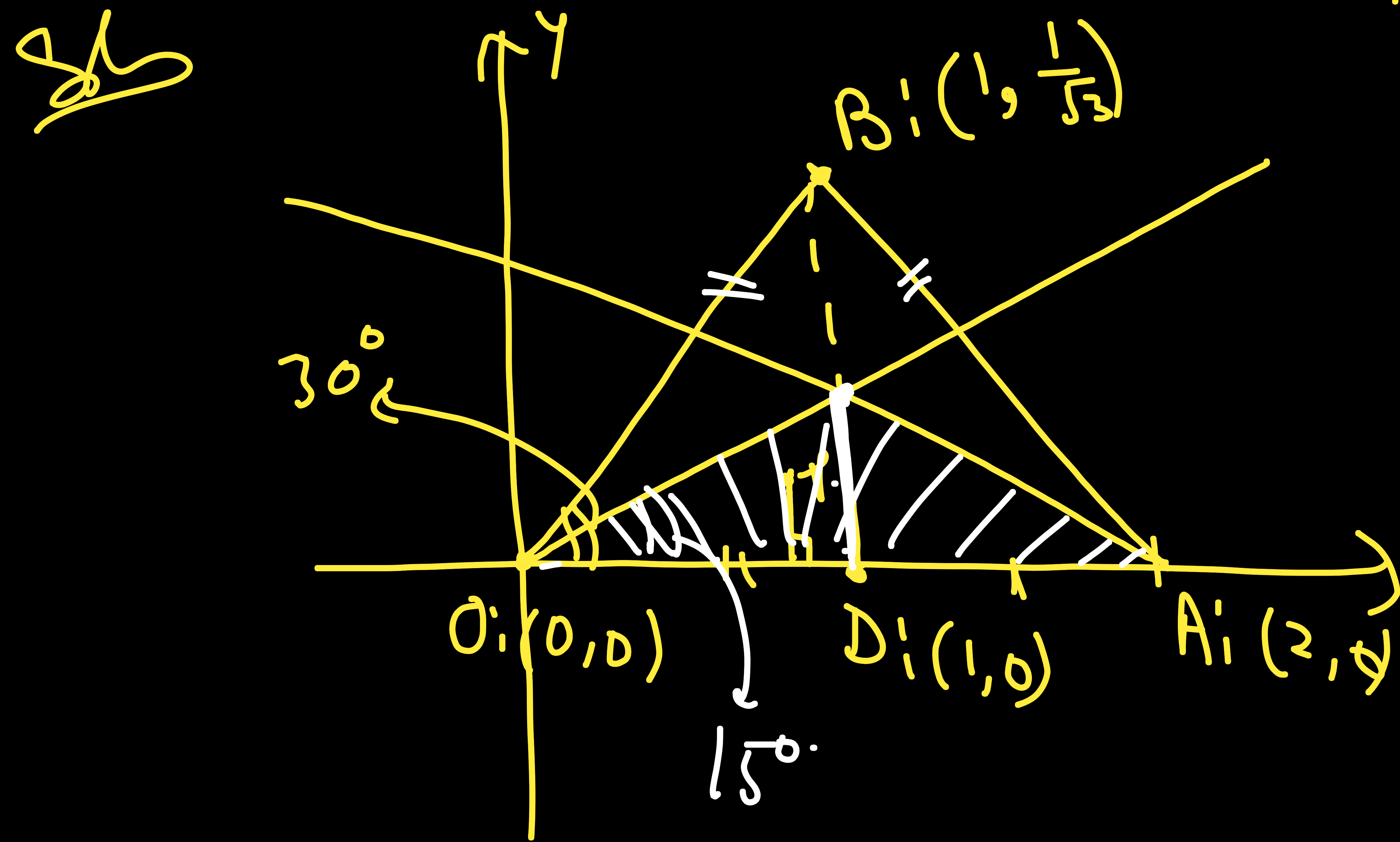
$$A'(k) = 0 = x_1' \cdot (kx_1 + 1 - (x_1^2 + 2x_1 - 3)) - x_2' \cdot (kx_2 + 1 - (x_2^2 + 2x_2 - 3)) + \int_{x_2}^{x_1} x dx$$

$$\Rightarrow \frac{x_1^2 - x_2^2}{2} = 0 \Rightarrow \frac{(x_1 - x_2)(x_1 + x_2)}{2} = 0$$

$$x_1 + x_2 = 0.$$
$$\Rightarrow x - z = 0 \Rightarrow \boxed{x = z}$$

Corresponding Area = D.I.X.

Ques Let $O: (0,0)$, $A: (2,0)$ & $B: (1, \frac{1}{\sqrt{3}})$ be the vertices of a triangle and let R be the region consisting of all those points P inside the $\triangle OAB$ such that $d(P, OA) \leq \min \{d(P, OB), d(P, AB)\}$ where $d(P, L)$ denotes distance of point P from line L . Find area of region R .



$$\checkmark \begin{aligned} d(P, OA) &\leq d(P, OB) \\ d(P, OA) &\leq d(P, AB) \end{aligned}$$

$$\begin{aligned} R. \text{Area} &= \frac{1}{2} \cdot 2 \cdot (1 \cdot \tan(15^\circ)) \\ &= \tan(15^\circ) \end{aligned}$$